



V2X-Assisted Pedestrian Safety System

A cooperative perception for urban pedestrian safety, developed in the CARLA simulation environment.

Roadside Simulator

CARLA SIMULATION

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Introduction: The Project

This document details the development of a **V2X (vehicle-to-everything) assistant pedestrian safety system**. Unlike traditional ADAS that relies solely on local onboard sensors, this project implements a **cooperative perception system** where a Roadside Unit (RSU) provides supplementary environmental data to the ego vehicle.

Traditional ADAS

Relies solely on local onboard sensors with no external data input.

V2X Cooperative System

A Roadside Unit (RSU) provides supplementary environmental data to the ego vehicle.

Motivations

Pedestrian safety remains a critical challenge in urban environments. The primary motivations for this project are:



Blind-Spot Reduction

Reduction of blind-spot accidents in complex urban intersections where local sensors fail to detect pedestrians.



Sensor Limitation Mitigation

Mitigation of sensor limitations (e.g., adverse weather) through collaborative RSU-to-vehicle communication that maintains detection capability when local vision is occluded.



Low-Latency V2X Validation

Validation of low-latency V2X architectures using the CARLA simulation environment to ensure realistic performance under network delay conditions.

System Requirements

Functional Requirements

1 Pedestrian Detection

Real-time identification of pedestrians using HOG descriptors.

2 V2X Communication

Receiving detection data from an external RSU camera.

3 Latency Management

Simulation of network delays via a queuing system to ensure realistic performance.

Non-Functional Requirements

Robustness

Operation consistency across varying weather presets in the CARLA simulation environment.

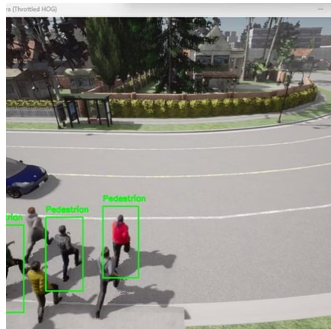
Efficiency

Lightweight implementation using OpenCV HOG to maintain high FPS in simulation.

Architecture: Sensors & Detection

The system utilizes two primary visual input streams. The **ego camera** provides a local perspective, while the **RSU camera** provides the global "bird's-eye" view.

$$< \frac{\text{width}}{\text{height}} <$$



We implemented the classical **OpenCV HOG pedestrian detector**. The HOG descriptor is initialized with a Support Vector Machine (SVM) detector:

```
hog = cv2.HOGDescriptor()  
hog.setSVMDetector(cv2.HOGDescriptor_getDefaultPeopleDetector())
```

An aspect ratio filter distinguishes pedestrians from vertical poles:

```
def is_valid_pedestrian(rect):  
    width, height = rect[2], rect[3]  
    ratio = width / height  
    return 0.2 < ratio < 0.6
```

V2X Latency Modeling

To simulate real-world network conditions, we implemented a **rolling buffer** that forces the system to handle delayed data packets. This models a 3-frame network delay, ensuring the ego-vehicle processes RSU data under realistic latency conditions:

```
import collections

# Simulation of a 3-frame network delay
v2x_latency_queue = collections.deque(maxlen=3)

def update_v2x_buffer(new_detection):
    v2x_latency_queue.append(new_detection)
    return v2x_latency_queue[0] # Return the oldest data
```



RSU Detects Pedestrian

Enters Rolling Buffer

Ego Vehicle Receives

The rolling buffer ensures the system processes RSU detection data under realistic network latency, simulating a 3-frame delay before the ego vehicle receives the packet.

Decision Logic: Emergency Braking

The emergency braking system follows a decision matrix that **prioritizes safety triggers**, with V2X warnings taking precedence over local sensor data:

```
def check_emergency_brake(is_warned_by_v2x, is_visible_locally, distance):  
    if is_warned_by_v2x:  
        return True # Immediate trigger from RSU  
    elif is_visible_locally or distance < 12.0:  
        return True # Emergency override  
    return False
```

1

V2X Warning

Immediate brake trigger if RSU reports pedestrian presence — highest priority.

2

Local Detection

Brake triggered if pedestrian is visible locally or distance is below 12.0 m.

3

No Action

The system remains passive when no threat is detected by either source.

Technologies Used



CARLA Simulator

Physics-based urban scenario replication for safe, repeatable testing of V2X scenarios.



OpenCV

Image processing, color space conversion (BGR2GRAY), and HOGDescriptor for pedestrian detection.



NumPy

Buffer handling and array operations for processing CARLA raw sensor data.



Testing & Results

Scenario Analysis

We compared baseline (unassisted) performance against V2X-assisted performance. Under **adverse weather (rain)**, the ego vehicle's local vision suffered from occlusion, whereas the RSU camera maintained a clear line of sight.

Detection Confidence

The use of `hit_threshold=0.3` provided the optimal balance between recall and precision for the pedestrian detector.



Collision Rate: Reduced in assisted mode compared to the baseline.

Conclusions & Future Work

Key Strength

The project demonstrated that V2X assistance provides a **significant safety margin**, proving that collaborative perception compensates for local sensor failure in complex urban scenarios.

Future Developments

01

Deep Learning Transition

Implementation of YOLOv8 for improved detection accuracy.

02

Dynamic Latency Model

Implementing a non-deterministic delay model for the V2X queue.

03

Sensor Fusion

Implementing a Kalman Filter to fuse ego-local and RSU-remote coordinate data.